Jeonghwan Song

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Research

Vehicle Trajectory Prediction, Motion Planning, Reinforcement Learning-Based Control

Education

Ulsan National Institute of Science and Technology, Ulsan, South Korea — Aug. 2022 – Now
Ph.D. in Artificial Intelligence
Advisor: Jeong hwan Jeon

Ulsan National Institute of Science and Technology, Ulsan, South Korea — Mar. 2016 – Aug. 2022*
B.S. in Electrical Engineering
Magna Cum Laude & Minor in Computer Science and Engineering

*Including Military Service, Republic of Korea Air Force, Mar. 2018 – Feb. 2020

Work Experiences

Robotics and Mobility Lab., Ulsan, South Korea — Aug. 2022 – Now
Graduate Research Assistant

RideFlux Inc., Jeju, South Korea — Sep. 2021 – Feb. 2022
Software Engineer Intern
Host: UNIST Coop Program

Teaching

Teaching Assistant at Ulsan National Institute of Science and Technology — Spring 2024, 2025
Automatic Control
Instructor: Jeong hwan Jeon

Teaching Assistant at Ulsan National Institute of Science and Technology — Jun. 2024 – Sep. 2024
LG Electronics DX Intensive Course PBL with **LG Electonics**
Instructor: Jeong hwan Jeon

Teaching Assistant at Ulsan National Institute of Science and Technology — Sep. 2023 – Dec. 2023
Nobatus Academia PBL (Project-Based Learning) with **ABH**
Instructor: Sunghoon Lim

Teaching Assistant at Ulsan National Institute of Science and Technology — Spring 2023
Introduction to AI Programming 1
Instructor: Michael Jon Burrell

International Conference

[1] **Jeonghwan Song** and Jeong hwan Jeon. "Decomposition-Based Intention-Aware Transformer for Vehicle Trajectory Prediction" In *IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, Oct., 2026.

[2] Sungjun Heo, Heedon Jeong, Eunchong Kim, Hyojae Lee, Jungeun Lee, Seongjae Lee, Minsu Joo, **Jeonghwan Song**, and Jeong hwan Jeon. "Capacity-Controlled Enforcement of Sparse Traffic-Light Cues for Rule-Conformant End-to-End Driving" In *IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, Oct., 2026.

[3] Dokeun Lee, **Jeonghwan Song**, Hyojae Lee, and Jeong hwan Jeon. "Attention-Based Reinforcement Learning for Context-Aware Path Tracking Control." In *IEEE Intelligent Vehicles Symposium (IV)*, June., 2026.

[4] Jaechan Shin, **Jeonghwan Song**, and Jeong hwan Jeon. "Efficient Vision-Based Driving Path Suggestion via Task-Specific Model Tuning." In *IEEE International Conference on Consumer Electronics Asia (ICCE-Asia)*, Oct., 2022.

International Journal

[1] **Jeonghwan Song*** et al. "Manuscript on urban motion planning for autonomous driving system (title withheld during review)" Co first-author submission, *under double-anonymous review.*

[2] Jaechan Shin, Hyeonwoo Jang, **Jeonghwan Song**, Seongoh Lee, Changsu Ha, Kyungdon Joo, and Jeong hwan Jeon. "SE3D: A Synthetic Stereo Event Camera Dataset for 3D Perception in Autonomous Driving Across Diverse Weather Conditions." Under review in *IEEE Transactions on Intelligent Vehicles (T-IV).*

Domestic Conference

[1] Changsu Ha, Hyeonwoo Jang, Jaechan Shin, **Jeonghwan Song**, and Jeong hwan Jeon. "EPL-SCI: Event Pseudo-LiDAR based 3D object detection for Suddenly Changing Illuminance." In *35th Workshop on Image Processing and Image Understanding (IPIU)*, Feb., 2023.

Domestic Patent

[1] Jeong hwan Jeon, Dokeun Lee, **Jeonghwan Song**. "Method and Apparatus for Vehicle Adaptive Control Based on Trajectory Uncertainty." In *Korean Patent*, patent pending.

[2] Jeong hwan Jeon, Changsu Ha, Hyeonwoo Jang, Jaechan Shin, **Jeonghwan Song**. "Method and Apparatus for Detecting 3D Object in Environment with Rapidly Changing Illuminance." In *Korean Patent No. 10-2952576*, registered Apr., 2026.

Awards and Honors

2nd place in 2026 Autonomous Driving AI Challenge (Future Trajectory Prediction) (KRW 2.5M) Nov. 2026
Host: Ministry of Science and ICT (MSIT), South Korea
Team: **Jeonghwan Song**, Kyeongmin Do, Kibeom Yoon, Seohee Lee

IEEE SMC 2026 Travel Grant (USD 600) Oct. 2026
IEEE Systems, Man, and Cybernetics Society (SMCS)

3rd place in 2025 Autonomous Driving AI Challenge (Future Trajectory Prediction) (KRW 1.5M) Nov. 2025
Host: Ministry of Science and ICT (MSIT), South Korea
Team: **Jeonghwan Song**, Kyeongmin Do, Dokeun Lee

Graduate Scholarship funded by UNIST Aug. 2022 – Now
Full-funded by Ulsan National Institute of Science and Technology, South Korea

Academic Performance Scholarship Mar. 2016 – Feb. 2022
Full-funded by Ulsan National Institute of Science and Technology, South Korea

National Science and Technology Scholarship Mar. 2016 – Feb. 2022
Full-funded by Ministry of Science and ICT (MSIT), South Korea

Open-Sourced Project

1. **SE3D**
   A Synthetic Stereo Event Camera Dataset for 3D Perception in Autonomous Driving Across Diverse Weather

Conditions.
https://github.com/rml-unist/se3d
Collaborator: Jaechan Shin, Hyeonwoo Jang, **Jeonghwan Song**